knowbridg...

KNOWBRIDGE AI

Never LetCriticalProjectKnowledgeLeave With People.

Intelligent Knowledge Handover Platform

Team: Code Crushers

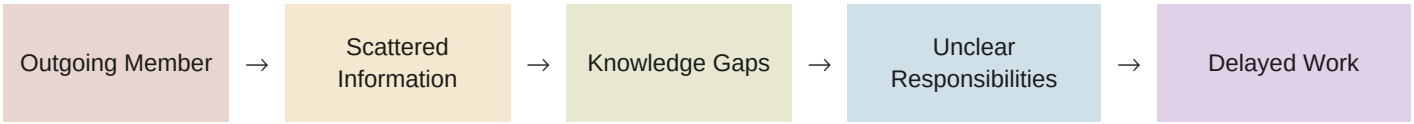
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Outgoing team member → Project knowledge → **Incoming team member**

THE HIDDEN PROBLEM OF KNOWLEDGE LOSS

- Important project information is scattered across documents, meetings, messages and task-boards.
- When a team member leaves, their experience and decision-making context may leave with them.
- Pending tasks and responsibilities become difficult to track for incoming members.
- New members spend time understanding old decisions; important dependencies can be missed.
- Manual handovers are inconsistent, incomplete, and impossible to audit.



INTRODUCING KNOWBRIDGE AI

1. Collect authorised project documents, meeting notes, and communication sources.
2. Analyse documents and project activities using AI and rule-based NLP.
3. Identify decisions, responsibilities and pending tasks automatically.
4. Highlight risks, dependencies and conflicting information.
5. Link important information to supporting evidence with source references.
6. Identify missing or conflicting information and flag it clearly.
7. Generate an organised, evidence-linked handover report (PDF and CSV).
8. Allow mandatory human review and approval before finalisation.
9. Enable recipient acknowledgement tracking with audit timestamps.

WITHOUT KnowBridge AI	WITH KnowBridge AI
• Scattered project information	• Organised evidence-linked report
• Unclear decision history	• Linked decisions, risks and tasks
• Manual handover efforts	• Mandatory human review and approval
• No audit trail	• Full audit log and version history
• Knowledge walks away with people	• Knowledge stays permanently

HOW THE SYSTEM IS DESIGNED

DATA INPUT	Project Documents · Meeting Notes · Task Boards · Code Repositories · Authorised Communication Sources
DATA PROCESSING	Permission Validation · Data Extraction · Text Pre-processing · Data Cleaning · Information Structuring
AI INTELLIGENCE	Google Gemini API · NLP Text Summarisation · Decision Extraction · Task & Ownership Detection · Risk & Dependency Identification · Rule-Based Fallback Engine
TRUST & VERIFICATION	Evidence Linking · Missing Info Detection · Contradiction Detection · Uncertainty Detection · Role-Based Access · Mandatory Human Approval
OUTPUT	Handover Report (PDF / CSV) · Decision Timeline · Pending Tasks · Risk Summary · Next Action Plan · Role-Specific Views · Acknowledgement Tracking

HOW KNOWBRIDGE AI WORKS

1. START	User logs in with their role — Admin, Project Manager, or Team Member.
2. UPLOAD	Upload authorised documents: PDF, DOCX, TXT, or CSV. Drag-and-drop is supported.
3. EXTRACT	System extracts text using PyMuPDF / pypdf / python-docx libraries automatically.
4. AI ANALYSIS	Google Gemini API analyses text — extracts decisions, tasks, risks, owners, deadlines and dependencies.
5. FALLBACK ENGINE	If Gemini API is unavailable, the built-in rule-based NLP engine activates automatically with zero downtime.
6. REVIEW QUEUE	All AI-extracted items enter a Human Review queue — not published directly to any report.
7. HUMAN APPROVAL	Reviewer inspects, edits, approves or rejects each item with reason and timestamp recorded.
8. REPORT	Approved items compile into a professional Handover Report. Download as PDF or export as CSV.
9. SECURE SHARING	Report shared with incoming members via role-based access control permissions.
10. ACKNOWLEDGEMENT	Recipient reads and formally acknowledges the handover. Timestamp recorded in audit log.
END	Knowledge is permanently preserved and searchable within the platform.

WHAT MAKES KNOWBRIDGE AI USEFUL?

Smart AI Knowledge Extraction	**Google Gemini API Integration**	**Rule-Based NLP Fallback Engine**
Decision Tracking	**Pending Task Identification**	**Risk & Dependency Detection**
Evidence-Linked Summaries	**Human Review & Approval Gate**	**Role-Based Access Control**
Handover Report (PDF & CSV)	**Acknowledgement Tracking**	**Project Timeline View**
Version History	**Audit Logs**	**Multi-Format Document Upload**
Interactive Analytics Dashboard	**Chart.js Visual Analytics**	**Drag-and-Drop File Upload**
Mobile-Responsive Design	**REST API Architecture**	**17 Functional Pages**
CSRF & Session Security		

TECHNOLOGIES BEHIND OUR PROJECT

Frontend	HTML5, CSS3, JavaScript (ES6+), Bootstrap 5, Chart.js, Lucide Icons, Responsive Layouts
Backend	Python 3, Flask, Flask-Login, Flask-WTF (CSRF), Flask REST API
Database	SQLite for local development · SQLAlchemy ORM · Designed for PostgreSQL migration
AI and NLP	Google Gemini API (gemini-1.5-flash) · Structured JSON schema output · Rule-based NLP fallback engine
Document Parsing	PyMuPDF / pypdf (PDF) · python-docx (Word) · Python CSV and JSON · Plain TXT
Report Generation	ReportLab (PDF download) · CSV export · Print-ready online report view
Security	CSRF protection · Session-based authentication · Backend-enforced RBAC · Permission checks on every API route
Deployment	Render.com · Gunicorn WSGI server · Environment-variable secrets (.env)
Visualisation	Dashboard with Chart.js · Project timeline · Risk distribution charts · Task status doughnut chart

ROLE-BASED ACCESS CONTROL

ROLE	PERMISSIONS
ADMIN	Manage all users and roles · Create / delete projects · View full audit logs · Approve / reject handovers · Manage system settings
PROJECT MANAGER	Manage assigned projects · Upload documents · Trigger AI analysis · Review and approve assigned handovers · Assign responsibilities
TEAM MEMBER	View authorised projects · Upload permitted documents · View assigned tasks · Acknowledge handovers · Update permitted task statuses

All permissions are enforced at the backend API level — hiding UI buttons is never the only security layer.

DASHBOARD AND APPLICATION FEATURES

Interactive Analytics Dashboard

All statistics are queried live from the database — no hardcoded numbers anywhere.

METRIC	DESCRIPTION
Total Projects	Count of all active projects in the workspace
Total Documents	All uploaded and processed documents
Extracted Decisions	AI-identified key project decisions
Pending Tasks	Open tasks awaiting completion or assignment
Identified Risks	Low / Medium / High risks flagged by AI and human reviewers
Completed Handovers	Successfully acknowledged handover reports
Awaiting Approvals	AI items pending human review and approval
Acknowledgement Status	Count of pending vs confirmed acknowledgements

Chart.js Visualisations Included

- Project progress bar chart (per-project completion percentage)
- Pending vs Completed tasks — doughnut chart
- Risk distribution by severity (Low / Medium / High) — pie chart
- Handover activity timeline chart
- Decision status overview (Pending / Approved / Rejected)

RELIABLE KNOWLEDGE HANDOVER

- Authorised source access — only permitted documents are analysed.
- Role-based permissions — each user sees only what they are allowed to see.
- Confidential information protection — uploads are never publicly accessible.
- Evidence-linked information — every extracted item links back to its source document.
- Missing and conflicting information alerts — flagged clearly, never silently ignored.
- Human verification before approval — no AI output reaches a report without human sign-off.
- Controlled information sharing — role-based report access and acknowledgement tracking.

Authorised Access	Data Processing	Permission Check	Evidence Validation	Human Review	Role-Based Sharing

WHERE CAN KNOWBRIDGE AI BE USED?

Corporate IT & Software Teams	Employee exit knowledge capture; onboarding acceleration for new developers.
College Project Teams	Smooth project handovers across academic batches or semesters.
Research Institutions	Preserve methodology, data decisions and open questions across researcher transitions.
Volunteer Organisations	Maintain operational continuity when volunteers rotate or leave.
Government & Public Sector	Formal documented handovers during departmental transitions with audit compliance.
Startups and SMEs	Prevent founder-dependency and single-point-of-failure knowledge risks.

WITHOUT KnowBridge AI	WITH KnowBridge AI
• Scattered information	• Organised evidence-linked report
• Manual searching and reading	• Instant AI-extracted summaries
• Unclear responsibilities	• Clear ownership and timelines
• Repeated explanations	• Better project continuity
• No audit trail	• Full version history and audit log

REST API ARCHITECTURE

KnowBridge AI exposes a full Flask REST API with JSON responses, used by both the web frontend and external integrations.

Authentication	POST /api/auth/register · POST /api/auth/login · POST /api/auth/logout
Projects	GET/POST /api/projects · GET/PUT/DELETE /api/projects/
Documents	POST /api/documents/upload · GET /api/documents · DELETE /api/documents/
AI Processing	POST /api/ai/analyze · POST /api/ai/summarize · POST /api/ai/extract-decisions · POST /api/ai/detect-risks
Tasks	GET/POST /api/tasks · PUT /api/tasks/
Reviews	GET /api/reviews · POST /api/reviews//approve · POST /api/reviews//reject
Reports	GET/POST /api/reports · GET /api/reports//download
Acknowledgements	GET /api/acknowledgements · POST /api/acknowledgements//confirm

WITHOUT KNOWBRIDGE AI

- Scattered information.
- Manual searching.
- Unclearresponsibilities.
- Repeated explanations.

WITH KNOWBRIDGE AI

- Organised project information.
- Evidence-linked summaries.
- Clear responsibilities.
- Better project continuity.

THANKYOU!!

Every project deserves a seamless continuation.

Team:Code Crushers







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